

Jay Prajapati

Calgary, AB | **Email:** jaykumararvindbhai.p@gmail.com | **Mobile:** (403) 953-4031 | [LinkedIn](#) | [Portfolio](#) | **EIT – APEGA**

Professional Summary

Mechanical Engineer in Training (EIT – APEGA) with a Master of Mechanical & Manufacturing Engineering (University of Calgary) and 1+ years of oil and gas engineering experience, specialising in FEA-based mechanical analysis, mechanical integrity assessment, piping stress engineering, and technical documentation across concurrent industrial client projects. At Zachry, the approach has always been to dig into the mechanics of why equipment behaves the way it does under load — not just report findings, but understand the physics driving them and translate that into clear, actionable engineering recommendations. Brings ANSYS, SolidWorks, Python, MATLAB, ASME and API code knowledge, ABSA AB-520 regulatory experience, a Lean Six Sigma Green Belt, and a track record of producing engineering deliverables that operations and integrity teams can act on.

Experience

Zachry Integrity Engineering Ltd. – Engineering Specialist

May 2025 – Present

Calgary-based engineering consultancy specialising in mechanical integrity, Fitness-for-Service assessment, and piping stress analysis for major oil and gas refinery and petrochemical clients across Western Canada.

- Performed FEA-based mechanical analysis and integrity assessments for pressure equipment and piping systems using ANSYS Workbench and APDL, evaluating structural behaviour, damage mechanisms, and failure risk under operating and transient loading conditions.
- Conducted API 579-1/ASME Level 3 Fitness-for-Service evaluations on refinery pressure vessels including a Vacuum Tower and Coker Main Fractionator, analysing post-WOL distortion, operating loads, and equipment geometry to support repair and continued operation decisions.
- Developed engineering design specifications and ASME B31.3-compliant calculations for piping components and pressure equipment, applying ASME BPVC VIII, B31J, PCC-1, and ABSA AB-520 codes to produce calculation packages accepted by client engineering teams.
- Performed root-cause analysis and failure mode investigations for pressure equipment and cyclically loaded piping, integrating thermography inspection data and operating history to diagnose damage mechanisms and develop corrective action recommendations.
- Coordinated technical deliverables across multi-discipline internal teams and client field personnel, managing concurrent project schedules and ensuring engineering outputs accurately reflected field conditions and operational constraints.

Circle K – Assistant Manager

Feb 2023 – Present

- Managed daily operations and a team of 8 staff, achieving a 14% sales increase, 20% reduction in safety incidents, and 15% decrease in stockouts through structured procedures, staff training, and an Excel VBA inventory management system.

Certifications

Revit for Mechanical and Plumbing Design Professional - [Certificate](#)

- Completed professional Revit MEP training covering HVAC, plumbing, and mechanical system modelling, BIM coordination, and engineering drawing production for commercial building projects.

Lean Six Sigma Green Belt - [Certificate](#)

- Completed structured continuous improvement, KPI development, performance monitoring, and data-driven problem solving methodology training.

Reading HVAC, Plumbing and other Drawings & Schematics - [Certificate](#)

- Completed formal P&ID, HVAC, plumbing, and engineering drawing interpretation training.

Mechanical Engineering Mastery Series: HVAC Engineering 101 - [Certificate](#)

- Completed HVAC engineering fundamentals including thermal load calculations, equipment selection, duct and hydronic system design, and energy efficiency considerations.

Advanced CREO Parametric Training for Product Design and Development - [Certificate](#)

- Completed advanced parametric modelling, assembly design, GD&T documentation, and manufacturing drawing production training.

Leadership & Projects

Differential Pressure Assessment of Catalyst Beds and Outlet Screens

- Assessed process equipment performance across catalyst beds and outlet screens using vendor drawings and operating data, modelling tray geometry and identifying loading effects to support equipment engineering decisions.

Bottom Flange Design of a Coke Drum and Mechanical Analysis

- Performed mechanical design and analysis for a coke drum bottom flange under cyclic operating and transient loads, conducting thickness sensitivity studies using thermography data to verify pressure integrity and support capital repair planning.

Anchor Bolt Vibration Analysis

- Assessed dynamic loading behaviour, resonance conditions, and long-term fatigue risk of anchor bolt connections on mechanical equipment foundations, producing reliability evaluations that informed maintenance planning and corrective action decisions.

Fitness-for-Service Assessment of Refinery Pressure Equipment

- Led mechanical integrity assessment for a Vacuum Tower and Coker Main Fractionator, performing API 579-1 FFS evaluations under ASME BPVC VIII and delivering repair and operating specifications accepted by client operations and integrity teams.

Diluent Tower WMBU and Post-WOL Distortion Assessment

- Modelled and analysed weld overlay repairs on a diluent tower in ANSYS to evaluate post-WOL distortion, local stress behaviour, and operating load effects, supporting repair planning and continued safe operation.

FEA Analysis of Coke Drum Bottom Nozzles

- Modelled and analysed coke drum bottom nozzles in ANSYS to evaluate local stress behaviour, flange rotation, and gasket seating stress under operating and design loads, supporting integrity assessment, leak-tightness review, and repair planning.

Minimum Thickness Evaluation of Vacuum Tower Cone Section

- Modelled and analysed a vacuum tower cone section in ANSYS to evaluate minimum required thickness using measured inspection data, operating conditions, and equipment geometry to support continued operation and repair planning.

SIFs Using FEA for Non-Standard Piping Components

- Developed ASME B31.3-compliant Stress Intensification Factors for non-standard jacketed pipe cross configurations using custom ANSYS FEA models, producing defensible design specifications where no standard B31J values existed.

Aerodynamic Analysis of Finite Span 3D Wing Performance using ANSYS, Python and MATLAB - [Project](#)

- Performed 3D CFD simulation and aerodynamic performance analysis using ANSYS, Python, and MATLAB, producing validated results through systematic mesh convergence studies and cross-method comparison.

CFD Analysis of NACA 4412 Airfoil with Grid Independence and Adaptive Meshing - [Project](#)

- Conducted systematic CFD simulation in ANSYS Fluent, achieving grid-converged aerodynamic performance data through mesh convergence studies and turbulence model validation.

3D CAD Model of Bench Vise Assembly – AutoCAD - [Project](#)

- Designed a detailed 3D bench vise assembly in AutoCAD with GD&T-annotated technical drawings, focusing on component precision, assembly constraints, and manufacturability.

3D CAD Model of Engine Blower – CREO - [Project](#)

- Developed a 3D engine blower model in CREO Parametric, focusing on precision parametric modelling and manufacturability, resulting in a 15% improvement in blower efficiency.

Energy-Efficient HVAC System Design for Retail Space Using Revit - [Project](#)

- Designed a complete HVAC system for a commercial retail building in Revit MEP, performing heating and cooling load calculations, equipment selection, and coordinated system layout under ASHRAE-referenced design methodology.

HVAC System Design for Multi-storey Office Using Revit

- Developed an energy-efficient HVAC system for a multi-zone, multi-storey office building in Revit MEP, addressing air distribution, load balancing, and system coordination in compliance with ASHRAE 90.1 and NEBC standards.

3D CAD Model of a 4-Cylinder Engine Assembly – SolidWorks - [Project](#)

- Designed a fully constrained 4-cylinder engine assembly in SolidWorks, producing fabrication-ready 2D drawings with

GD&T callouts through precision parametric modelling, assembly constraints, and manufacturability-focused design.

HAZOP Analysis of Sulphur Recovery Unit for Process Safety and Risk Mitigation - [Project](#)

- Conducted systematic process hazard identification and risk assessment for a Sulphur Recovery Unit, documenting hazard scenarios, integrity threats, and corrective actions in a formal HAZOP report.

Random Vibrational Analysis of a Shock Absorber - [Project](#)

- Developed a 3D solid-element FEA model of a shock absorber assembly in ANSYS, applying PSD acceleration loading across 5–50 Hz with local mesh refinement to evaluate directional deformation and equivalent von Mises stress under random vibration conditions.

Education

Schulich School of Engineering, University of Calgary – Master of Mechanical & Manufacturing Engineering

Gujarat Technological University – Bachelor of Aeronautical Engineering

Courses: Finite Element Analysis, Computational Fluid Dynamics, Thermodynamics, Engineering Design, Advanced Material Engineering, Project Management, Aerodynamics, Manufacturing Optimization

Concentration: Mechanical Design, Engineering Analysis, Mechanical Integrity, Pressure Equipment Design, Stress Analysis

Skills

Technical Skills: FEA, Mechanical Integrity Evaluation, API 579-1 FFS, ASME BPVC VIII, ASME B31.3, ASME B31J, ASME PCC-1 Appendix O, ABSA AB-520, Piping Stress Analysis, Root-Cause Analysis, Failure Analysis, Vibration and Dynamic Analysis, Random Vibration Analysis, CFD, HVAC System Design, GD&T, Material Selection, Engineering Calculations, P&ID Reading, HAZOP Participation, Technical Documentation

Software Knowledge: ANSYS Workbench & APDL, SolidWorks, AutoCAD, Revit MEP, CREO Parametric, CATIA V5, ABAQUS, MATLAB, Python, Microsoft Excel (Advanced, VBA), Microsoft Office

Soft Skills: Analytical Problem-Solving, Technical Communication, Multi-discipline Coordination, Attention to Detail, Continuous Improvement, Independent Work, Safety-First Mindset, Valid Class 5 Driver's Licence